



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 10

Name: Premarubha N

Reg No: 192525020

Program:

```
def insertion_sort(arr):  
    for i in range(1, len(arr)):  
        key = arr[i]  
        j = i - 1  
        while j >= 0 and arr[j] > key:  
            arr[j + 1] = arr[j]  
            j -= 1  
        arr[j + 1] = key  
    return arr  
  
arr = [12, 11, 13, 5, 6]  
  
print("Sorted array:", insertion_sort(arr))
```

Output:

The screenshot shows a code editor with a dark theme. The file name is 'main.py'. There are buttons for 'Visualize' and 'Run'. The code is as follows:

```
1 def insertion_sort(arr):  
2     for i in range(1, len(arr)):  
3         key = arr[i]  
4         j = i - 1  
5  
6         while j >= 0 and arr[j] > key:  
7             arr[j + 1] = arr[j]  
8             j -= 1  
9  
10        arr[j + 1] = key  
11  
12    return arr  
13  
14  
15 arr = [12, 11, 13, 5, 6]  
16  
17 print("Sorted array:", insertion_sort(arr))
```

The output panel on the right shows: Sorted array: [5, 6, 11, 12, 13]